WealthWise

Database Schema Documentation

Mutual Fund & SIP Tracking Platform | Internship Project 2026

1. Overview

This document describes the complete MySQL database schema for the WealthWise application — a mutual fund and SIP tracking platform built using Java Spring Boot, MySQL, and React.

#	Table Name	Purpose
1	users	Authentication for Admin and Investor. Stores login credentials, role, status and admin key.
2	amc	Asset Management Companies — the fund houses that create and manage mutual funds.
3	mutual_funds	All mutual funds listed on the platform including NAV, category, risk and scheme code.
4	nav_history	Daily NAV records fetched from MFAPI for each mutual fund.
5	investments	Every investment transaction made by an investor (SIP installment or lumpsum).
6	sips	SIP plans created by investors — stores frequency, dates, amount and status.
7	sip_transactions	Monthly SIP transaction records — tracks PAID, MISSED and PENDING status per installment.
8	portfolio	Summary of each investor's holdings per fund — total units and total invested amount.
9	alerts	In-app notifications sent by admin to investors — automated and manual alerts.

2. Table Relationships

Parent Table	Child Table	Relationship	Constraint
amc	mutual_funds	One AMC → Many Funds	ON DELETE RESTRICT
mutual_funds	nav_history	One Fund → Many NAV records	ON DELETE CASCADE
mutual_funds	investments	One Fund → Many Investments	ON DELETE RESTRICT
mutual_funds	sips	One Fund → Many SIPs	ON DELETE RESTRICT
mutual_funds	sip_transactions	One Fund → Many Transactions	ON DELETE RESTRICT
mutual_funds	portfolio	One Fund → Many Portfolio rows	ON DELETE RESTRICT
users	investments	One User → Many Investments	ON DELETE CASCADE
users	sips	One User → Many SIPs	ON DELETE CASCADE
users	sip_transactions	One User → Many Transactions	ON DELETE CASCADE
users	portfolio	One User → Many Portfolio rows	ON DELETE CASCADE
users	alerts (sent_to)	One User → Many Alerts received	ON DELETE CASCADE
users	alerts (sent_by)	Admin → Many Alerts sent	ON DELETE RESTRICT
sips	sip_transactions	One SIP → Many Monthly Transactions	ON DELETE CASCADE

3. Calculated Fields (Never Stored in DB)

The following values are always computed in the Spring Boot backend at runtime and never stored in the database:

Field	Formula	Used On Page
current_value	total_units × current_nav	Portfolio, Dashboard
profit_loss	current_value − total_invested	Portfolio, Dashboard
return %	(profit_loss ÷ total_invested) × 100	Portfolio, Dashboard
units_purchased	amount ÷ nav_at_investment	Add Investment
total_aum	SUM of all portfolio.total_invested	Admin Dashboard
next_due_date	previous_due_date + frequency (1 month / 3 months)	SIPs, Dashboard

4. Table Details

Table 1 — users

Purpose: Stores authentication data for both Admin and Investor roles.

Column	Data Type	Constraints	Notes
user_id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for every user
name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(100)	UNIQUE, NULLABLE	Required for investors. NULL for admin.

Column	Data Type	Constraints	Notes
phone	VARCHAR(15)	UNIQUE, NULLABLE	Required for investors. NULL for admin.
password	VARCHAR(255)	NULLABLE	BCrypt hashed. Required for investors.
role	ENUM	NOT NULL, DEFAULT 'INVESTOR'	Values: ADMIN or INVESTOR
status	ENUM	NOT NULL, DEFAULT 'ACTIVE'	Values: ACTIVE or INACTIVE
admin_key	VARCHAR(100)	UNIQUE, NULLABLE	Secret key for admin login only. Inserted once manually.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date

Admin Insert (run once manually in MySQL):

```
INSERT INTO users (name, role, admin_key) VALUES ('Admin', 'ADMIN', 'ADMIN@2026');
```

Table 2 — amc

Purpose: Stores Asset Management Companies added by the admin.

Column	Data Type	Constraints	Notes
amc_id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique AMC identifier
amc_name	VARCHAR(150)	NOT NULL, UNIQUE	e.g. SBI Mutual Fund, HDFC Mutual Fund
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date AMC was added

Table 3 — mutual_funds

Purpose: All mutual funds listed on the platform by the admin.

Column	Data Type	Constraints	Notes
fund_id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique fund identifier
fund_name	VARCHAR(200)	NOT NULL, UNIQUE	e.g. SBI Bluechip Fund
category	ENUM	NOT NULL	Values: EQUITY, DEBT, HYBRID
risk_level	ENUM	NOT NULL	Values: LOW, MEDIUM, HIGH
current_nav	DECIMAL(10,4)	NOT NULL, DEFAULT 0.0000	Updated daily via MFAPI scheduled job
scheme_code	INT	UNIQUE, NULLABLE	MFAPI scheme code — used to fetch live NAV
amc_id	BIGINT	FOREIGN KEY → amc(amc_id)	ON DELETE RESTRICT
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date fund was added

Table 4 — nav_history

Purpose: Stores daily NAV records per fund fetched from MFAPI.

Column	Data Type	Constraints	Notes
nav_id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique record identifier
fund_id	BIGINT	FOREIGN KEY → mutual_funds(fund_id)	ON DELETE CASCADE
nav_price	DECIMAL(10,4)	NOT NULL	NAV value on that date
nav_date	DATE	NOT NULL	Date of the NAV record
(fund_id, nav_date)	—	UNIQUE CONSTRAINT	Prevents duplicate NAV entry per fund per day

Table 5 — investments

Purpose: Records every investment transaction — both SIP installments and lumpsum purchases.

Column	Data Type	Constraints	Notes
investment_id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique transaction identifier
user_id	BIGINT	FOREIGN KEY → users(user_id)	ON DELETE CASCADE
fund_id	BIGINT	FOREIGN KEY → mutual_funds(fund_id)	ON DELETE RESTRICT
investment_type	ENUM	NOT NULL	Values: SIP or LUMPSUM
amount	DECIMAL(12,2)	NOT NULL, CHECK > 0	Amount invested in this transaction
nav_at_investment	DECIMAL(10,4)	NOT NULL	NAV price on the day of investment (historical record)
units_purchased	DECIMAL(12,4)	NOT NULL, CHECK > 0	Calculated: amount ÷ nav_at_investment
investment_date	DATE	NOT NULL	Date of the transaction
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

Table 6 — sips

Purpose: Stores SIP plans created by investors. One row per SIP plan.

Column	Data Type	Constraints	Notes
sip_id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique SIP plan identifier
user_id	BIGINT	FOREIGN KEY → users(user_id)	ON DELETE CASCADE
fund_id	BIGINT	FOREIGN KEY → mutual_funds(fund_id)	ON DELETE RESTRICT

Column	Data Type	Constraints	Notes
amount	DECIMAL(12,2)	NOT NULL, CHECK > 0	Monthly SIP amount
frequency	ENUM	NOT NULL, DEFAULT 'MONTHLY'	Values: MONTHLY or QUARTERLY
start_date	DATE	NOT NULL	Date the SIP begins
next_due_date	DATE	NOT NULL	Updated after each successful payment
end_date	DATE	NULLABLE	NULL for open-ended SIPs. CHECK end_date > start_date.
status	ENUM	NOT NULL, DEFAULT 'ACTIVE'	Values: ACTIVE, PAUSED, COMPLETED
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	SIP creation date

Table 7 — sip_transactions

Purpose: Monthly installment records per SIP. Powers SIP History and Missed tabs on My SIPs page.

Column	Data Type	Constraints	Notes
txn_id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique transaction record
sip_id	BIGINT	FOREIGN KEY → sips(sip_id)	ON DELETE CASCADE
user_id	BIGINT	FOREIGN KEY → users(user_id)	ON DELETE CASCADE
fund_id	BIGINT	FOREIGN KEY → mutual_funds(fund_id)	ON DELETE RESTRICT
amount	DECIMAL(12,2)	NOT NULL, CHECK > 0	SIP amount for this installment
due_date	DATE	NOT NULL	Scheduled payment date
paid_date	DATE	NULLABLE	Actual payment date. NULL if not paid.

Column	Data Type	Constraints	Notes
status	ENUM	NOT NULL, DEFAULT 'PENDING'	Values: PAID, MISSED, PENDING
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

Status Logic:

PENDING → SIP due date has not yet arrived

PAID → Investor paid on time (paid_date is set)

MISSED → Due date has passed and no payment was made

Table 8 — portfolio

Purpose: Summary of each investor's holdings per fund. Updated after every investment.

Column	Data Type	Constraints	Notes
portfolio_id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique portfolio record
user_id	BIGINT	FOREIGN KEY → users(user_id)	ON DELETE CASCADE
fund_id	BIGINT	FOREIGN KEY → mutual_funds(fund_id)	ON DELETE RESTRICT
total_units	DECIMAL(14,4)	NOT NULL, DEFAULT 0.0000	Accumulates with every investment
total_invested	DECIMAL(14,2)	NOT NULL, DEFAULT 0.00	Sum of all amounts invested in this fund
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	First investment date
updated_at	TIMESTAMP	ON UPDATE CURRENT_TIMESTAMP	Last updated automatically
(user_id, fund_id)	—	UNIQUE CONSTRAINT	One portfolio row per investor per fund

Table 9 — alerts

Purpose: In-app notifications sent by admin to investors. One row per investor per alert.

Column	Data Type	Constraints	Notes
alert_id	BIGINT	PRIMARY KEY, AUTO_INCREMENT	Unique alert identifier
sent_by	BIGINT	FOREIGN KEY → users(user_id)	ON DELETE RESTRICT — admin user_id
sent_to	BIGINT	FOREIGN KEY → users(user_id)	ON DELETE CASCADE — investor user_id
alert_type	ENUM	NOT NULL	Values: SIP_DUE, MISSED_SIP, SIP_COMPLETION, GENERAL
message	TEXT	NOT NULL	Final personalized message after template is filled
is_read	BOOLEAN	NOT NULL, DEFAULT FALSE	Updated to TRUE when investor reads the alert
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date alert was sent

Alert Templates (stored in Java backend, not in DB):

```
SIP_DUE      → "Dear {name}, your SIP of Rs.{amount} for {fund} is due on {date}."
MISSED_SIP   → "Dear {name}, your SIP of Rs.{amount} for {fund} on {date} was missed."
SIP_COMPLETION → "Dear {name}, your SIP for {fund} is complete. Total invested: Rs.{amount}."
```

5. Key Constraints Summary

Constraint Type	Table.Column	Rule
PRIMARY KEY	All tables — *_id column	Auto increment unique identifier
UNIQUE	users.email	No two investors can share an email
UNIQUE	users.phone	No two investors can share a phone number
UNIQUE	users.admin_key	Only one admin key exists in the system
UNIQUE	amc.amc_name	No duplicate AMC names
UNIQUE	mutual_funds.fund_name	No duplicate fund names
UNIQUE	mutual_funds.scheme_code	Each MFAPI scheme code is unique
UNIQUE	portfolio (user_id, fund_id)	One portfolio summary row per investor per fund
UNIQUE	nav_history (fund_id, nav_date)	One NAV record per fund per day
FK + RESTRICT	mutual_funds → amc	Cannot delete AMC if it has funds linked
FK + RESTRICT	investments → mutual_funds	Cannot delete fund if it has investments
FK + CASCADE	investments → users	Deleting user removes all their investments
FK + CASCADE	sips → users	Deleting user removes all their SIPs
FK + CASCADE	sip_transactions → sips	Deleting SIP removes its transaction records
FK + CASCADE	portfolio → users	Deleting user removes their portfolio
FK + CASCADE	alerts → sent_to (users)	Deleting user removes their notifications
CHECK	investments.amount > 0	Investment amount must be positive
CHECK	sips.amount > 0	SIP amount must be positive
CHECK	sips.end_date > start_date	SIP end date must be after start date

6. MFAPI Integration

Live NAV data is fetched from the free MFAPI service (https://api.mfapi.in) using a Spring Boot scheduled job.

Step	Description
1	Admin adds a mutual fund and enters the MFAPI scheme_code (found at https://api.mfapi.in/mf/search?q=fund+name)
2	Spring Boot @Scheduled job runs daily at 8 PM
3	For each fund, backend calls https://api.mfapi.in/mf/{scheme_code}
4	Response returns latest NAV and date
5	current_nav is updated in mutual_funds table
6	A new row is inserted in nav_history table with today's NAV and date
7	Portfolio current_value is recalculated at runtime using updated current_nav